Quantum Risk Assessment

Executive Security Report

Scan ID:	**scan_6ba74907**
Scan Target:	**N/A**
Date:	**2026-07-30 21:22:54**

Executive Summary

Grade: F

Overall Score: 0 / 100

Findings Summary:

7 Critical **3 High** **0 Medium** **0 Low**

Migration Impact Estimate

Estimated Storage Bloat	33.0696 MB/day
Handshake Latency Penalty	+5.04 ms per connection

Bandwidth Increase	788.8%

Note: Assumes 1000 ops/day. Sourced from local hardware benchmarks.

Detailed Findings & Analysis

ECC (256 bits) Critical

Source: E:\QuantumSafe Sentinel\backend\tests\..\..\sample-data\certs\ecc_p256_kex.pem

Context & Usage: production_cert / key_exchange

Stage 1 (Hybrid Mode): `X25519+Kyber768`

NIST SP 800-208 / hybrid-mode guidance recommends pairing classical and PQC algorithms during the transition period to maintain FIPS compliance while providing quantum resistance.

Stage 2 (Pure PQC): `ML-KEM (Kyber)`

RSA (2048 bits) Critical

Source: E:\QuantumSafe Sentinel\backend\tests\..\..\sample-data\certs\rsa_2048_sign.pem

Context & Usage: production_cert / signing

Stage 1 (Hybrid Mode): `RSA3072+ML-DSA-65`

NIST SP 800-208 / hybrid-mode guidance recommends pairing classical and PQC algorithms during the transition period to maintain FIPS compliance while providing quantum resistance.

Stage 2 (Pure PQC): `ML-DSA (Dilithium)`

RSA (4096 bits) High

Source: E:\QuantumSafe Sentinel\backend\tests\..\..\sample-data\certs\rsa_4096_misc.pem

Context & Usage: production_cert / signing

Stage 1 (Hybrid Mode): `RSA3072+ML-DSA-65`

NIST SP 800-208 / hybrid-mode guidance recommends pairing classical and PQC algorithms during the transition period to maintain FIPS compliance while providing quantum resistance.

Stage 2 (Pure PQC): `ML-DSA (Dilithium)`

RSA (2048 bits) High

Source: E:\QuantumSafe Sentinel\backend\tests\..\..\sample-data\vulnerable-repo\auth.py (Line 5)

Context & Usage: code / misc

Stage 1 (Hybrid Mode): `X25519+Kyber768 or RSA3072+ML-DSA-65`

NIST SP 800-208 / hybrid-mode guidance recommends pairing classical and PQC algorithms during the transition period to maintain FIPS compliance while providing quantum resistance.

Stage 2 (Pure PQC): `ML-KEM or ML-DSA (Usage Unclear - Verify)`

ECC (256 bits) Critical

Source: E:\QuantumSafe Sentinel\backend\tests\..\..\sample-data\vulnerable-repo\tls_helper.py (Line 6)

Context & Usage: code / key_exchange

Stage 1 (Hybrid Mode): `X25519+Kyber768`

NIST SP 800-208 / hybrid-mode guidance recommends pairing classical and PQC algorithms during the transition period to maintain FIPS compliance while providing quantum resistance.

Stage 2 (Pure PQC): `ML-KEM (Kyber)`

RSA (Unknown bits) High

Source: E:\QuantumSafe Sentinel\backend\tests\..\..\sample-data\configs\nginx.conf (Line 9)

Context & Usage: config / signing

Stage 1 (Hybrid Mode): `RSA3072+ML-DSA-65`

NIST SP 800-208 / hybrid-mode guidance recommends pairing classical and PQC algorithms during the transition period to maintain FIPS compliance while providing quantum resistance.

Stage 2 (Pure PQC): `ML-DSA (Dilithium)`

DH (Unknown bits) Critical

Source: E:\QuantumSafe Sentinel\backend\tests\..\..\sample-data\configs\nginx.conf (Line 9)

Context & Usage: config / key_exchange

Stage 1 (Hybrid Mode): `X25519+Kyber768`

NIST SP 800-208 / hybrid-mode guidance recommends pairing classical and PQC algorithms during the transition period to maintain FIPS compliance while providing quantum resistance.

Stage 2 (Pure PQC): `ML-KEM (Kyber)`

ECDH (Unknown bits) Critical

Source: E:\QuantumSafe Sentinel\backend\tests\..\..\sample-data\configs\nginx.conf (Line 9)

Context & Usage: config / key_exchange

Stage 1 (Hybrid Mode): `X25519+Kyber768`

NIST SP 800-208 / hybrid-mode guidance recommends pairing classical and PQC algorithms during the transition period to maintain FIPS compliance while providing quantum resistance.

Stage 2 (Pure PQC): `ML-KEM (Kyber)`

ECDH (Unknown bits) Critical

Source: E:\QuantumSafe Sentinel\backend\tests\..\..\sample-data\configs\sshd_config (Line 6)

Context & Usage: config / key_exchange

Stage 1 (Hybrid Mode): `X25519+Kyber768`

NIST SP 800-208 / hybrid-mode guidance recommends pairing classical and PQC algorithms during the transition period to maintain FIPS compliance while providing quantum resistance.

Stage 2 (Pure PQC): `ML-KEM (Kyber)`

DH (Unknown bits) Critical

Source: E:\QuantumSafe Sentinel\backend\tests\..\..\sample-data\configs\sshd_config (Line 6)

Context & Usage: config / key_exchange

Stage 1 (Hybrid Mode): `X25519+Kyber768`

> *NIST SP 800-208 / hybrid-mode guidance recommends pairing classical and PQC algorithms during the transition period to maintain FIPS compliance while providing quantum resistance.*

> **Stage 2 (Pure PQC):** `ML-KEM (Kyber)`